

EY_rev_test

Section 1 - Coding

Problem - A

Find the B-th Positive Integer Not Divisible by A

Problem Description

Given two positive integers A and B, find the B-th smallest positive integer that is not divisible by A.

Input Format

* A single line containing two space-separated integers, A and B.

Output Format

* Print a single integer representing the B-th positive integer that is not divisible by A.

Constraints

- $2 \leq A \leq 10^9$
- $1 \leq B \leq 10^9$

Example 1

Input:

3 7

Output:

10

Example 2

Input:

2 5

Output:

9

Problem - B

Coloured Zenga

You are given a stack of blocks, where each block has a color represented by a character. In one move, you can remove two or more adjacent blocks of the same color.

When blocks are removed, the remaining blocks close the gap and form a new sequence.

Your task is to determine the minimum number of moves required to remove all the blocks.

Input:

- A string **S** representing the sequence of colored blocks.
- Each character represents the color of a block.

Output:

- Print the minimum number of moves required to remove all blocks.
- If it is impossible to remove all blocks, return **-1** (if applicable to the given version).

Example 1

Input:

AABBBAA

Output:

2

Example 2

Input:

AABBBCC

Output:

2

Problem - C**Iron Magnet**

You are given a string consisting only of the characters **N** and **S**, representing the north and south poles of magnets.

The magnets need to be arranged in a required alternating pattern:

NSNSNS...

or

SNSNSN...

In one operation, you can swap two positions in the string.

Find the minimum number of swaps required to make the string follow an alternating **N-S** pattern.

If it is impossible to form such a pattern, return **-1**.

Input:

- A string **S** containing only **N** and **S**.

Output:

- The minimum number of swaps required.
- Return **-1** if no valid alternating arrangement is possible.

Example 1:

Input:

S = "NNSS"

Output:

1

Example 2:

Input:

S = "NNNSS"

Output:

-1

Problem - D**Maximum Loss in Stock Prices**

You are given an array **P** representing the price of a stock on consecutive days.

You must choose one day to buy the stock and a later day to sell it.

Unlike the usual maximum-profit problem, here you need to find the maximum possible loss.

The loss is calculated as:

Loss = Buying Price - Selling Price

where the selling day must occur after the buying day.

Return the maximum possible loss.

Input:

- An integer **N**, representing the number of days.
- An array **P** of **N** integers representing stock prices.

Output:

- Print the maximum possible loss.

Example 1:

Input:

N = 6

P = [10, 7, 8, 5, 11, 3]

Output:

8

Example 2:

Input:

N = 5

P = [10, 9, 7, 6, 5]

Output:

5

Section 2 - Quantitative Aptitude

- 1) A pizza restaurant wants to offer its food via delivery. The pizzas can stay at the optimal temperature for no longer than 20 minutes. The speed limit in the city is 30 miles per hour, what is the maximum delivery radius to ensure the pizzas arrive at the optimal temperature?
- 2) The number $234X98$ is divisible by 3. What is the sum of the maximum and minimum possible value of x ?
- 3) Ayishree invested RS. 200,000 at 10% per annum. Interest is compounded half yearly. What is the amount and interest collected after 1.5 years?
- 4) Three cars started from Kolkata for patna reaches in 10 hrs, 15 hours and 12 hrs respectively. What is the ration of their speeds.
- 5) A car is bought for RS. 800,00 and company laws allow it to be depreciated @20% annum. It is sold after 4 years for 400,000. What is gain/loss?
- 6) If $X=0.59$ and $Y=1.41$, then find the value of
 - a) $x^2+2xy+y^2$
 - b) $x^3+3xy^2+y^3$

- 7) What is the discount offered when a shopkeeper offers "Buy 2 Get 1 Free"?
- 8) A number is supposed to be multiplied by $\frac{2}{3}$ but by mistake, got multiplied by $\frac{3}{2}$. What is the percentage increase/decrease in the value which was supposed to be there?
- 9) How many zeros are there at the end of the expression: $1*2*3*...*500$?
- 10) Xyz is a number, when divided by 54 leaves the remainder 3. What is the remainder when it is divided by 18?

Section 3 - Logical Reasoning

- 1) Find the next number in the series: 2, 6, 12, 20, 30, ?
- 2) Find the next term in the series: Z, W, T, Q, ?
- 3) In a certain code, BOOK is written as CPPL. How is DOOR written?
- 4) What will come next in the series: 3, 9, 27, 81, ?
- 5) All roses are flowers. Some flowers are red. Which statement is definitely true?
 - A) All roses are red
 - B) Some roses are red
 - C) All roses are flowers
 - D) No roses are red
- 6) Pointing to a woman, Anuj says, "She is the daughter of my grandfather's only son." How is the woman related to Anuj?
- 7) Find the missing number: 2, 5, 10, 17, 26, ?
- 8) If CAT is coded as DBU, how is DOG coded using the same pattern?
- 9) A person walks 5 km north, then turns right and walks 3 km. He then turns right again and walks 5 km. In which direction is he from his starting point?
- 10) If all managers are employees and some employees are graduates, which of the following is definitely true?
 - A) All managers are graduates
 - B) Some managers are graduates
 - C) All managers are employees
 - D) No managers are graduates

Answers:**Section 2 - Quantitative Aptitude**

1. Answer: 10 miles

Explanation: Speed = 30 miles/hour. Time = 20 minutes = $\frac{1}{3}$ hour. Distance = Speed $\times$ Time = $30 \times \frac{1}{3} = 10$ miles. Therefore, the maximum delivery radius is 10 miles.

2. Answer: 8

Explanation: For a number to be divisible by 3, the sum of its digits must be divisible by 3. For 234X98, the digit sum is $2 + 3 + 4 + X + 9 + 8 = 26 + X$. The possible values of X are 1, 4, and 7. Minimum X = 1 and maximum X = 7. Their sum = $1 + 7 = 8$.

3. Answer: Amount = ₹2,31,525 and Interest = ₹31,525

Explanation: Principal = ₹2,00,000, rate = 10% per annum, compounded half-yearly, and time = 1.5 years = 3 half-years. Rate per half-year = $10\% \div 2 = 5\%$. Amount = $2,00,000 \times (1.05)^3 = ₹2,31,525$. Interest = $₹2,31,525 - ₹2,00,000 = ₹31,525$.

4. Answer: 6:4:5

Explanation: For the same distance, speed is inversely proportional to time. The times are 10, 15, and 12 hours. Therefore, speed ratio = $\frac{1}{10} : \frac{1}{15} : \frac{1}{12}$. Taking LCM 60, we get 6:4:5. Therefore, the ratio of their speeds is 6:4:5.

5. Answer: Gain = ₹72,320

Explanation: Assuming the purchase price is ₹8,00,000, depreciation is 20% per year. After 4 years, value = $8,00,000 \times (80/100)^4 = ₹3,27,680$. Selling price = ₹4,00,000. Gain = $₹4,00,000 - ₹3,27,680 = ₹72,320$.

- 6a) Answer: 4

Explanation: $x^2 + 2xy + y^2 = (x + y)^2$. Given $x = 0.59$ and $y = 1.41$, $x + y = 2$.

Therefore, $(x + y)^2 = 2^2 = 4$.

- 6b) Answer: 6.527537

Explanation: Given $x = 0.59$ and $y = 1.41$. $x^3 = 0.205379$, $3xy^2 = 3 \times 0.59 \times 1.9881 = 3.518937$, and $y^3 = 2.803221$. Therefore, $x^3 + 3xy^2 + y^3 = 0.205379 + 3.518937 + 2.803221 = 6.527537$.

7. Answer: 33.33%

Explanation: Suppose each item costs ₹100. The normal cost of 3 items is ₹300. Under "Buy 2 Get 1 Free", you pay ₹200 for 3 items. Discount = ₹300 – ₹200 = ₹100. Discount percentage = $(100/300) \times 100 = 33.33\%$.

8. Answer: 125% increase

Explanation: Suppose the number is 1. The correct value is $1 \times \frac{2}{3} = \frac{2}{3}$. The incorrect value is $1 \times \frac{3}{2} = \frac{3}{2}$. Increase = $\frac{3}{2} - \frac{2}{3} = \frac{5}{6}$. Percentage increase = $(\frac{5}{6}) \div (\frac{2}{3}) \times 100 = 125\%$. Therefore, there is a 125% increase.

9. Answer: 124 zeros

Explanation: The expression $1 \times 2 \times 3 \times \dots \times 500$ is 500!. Trailing zeros are formed by factors of $10 = 2 \times 5$. Since there are more factors of 2 than 5, we count the factors of 5. Number of factors of 5 = $500/5 + 500/25 + 500/125 = 100 + 20 + 4 = 124$. Therefore, there are 124 zeros.

10. Answer: 3

Explanation: Let the number be N. Since N leaves remainder 3 when divided by 54, $N = 54k + 3$. Since 54 is divisible by 18, $54k$ is also divisible by 18. Therefore, when N is divided by 18, the remainder is also 3.

Answers:

Section 2 - Logical Reasoning

- 1) Answer: 42

Explanation: The differences are 4, 6, 8, and 10. The difference increases by 2 each time, so the next difference is 12. Therefore, $30 + 12 = 42$.

- 2) Answer: N

Explanation: Each letter moves 3 positions backward in the alphabet. $Z \rightarrow W \rightarrow T \rightarrow Q \rightarrow N$. Therefore, the answer is N.

- 3) Answer: EPPS

Explanation: Each letter is shifted forward by one position. $B \rightarrow C, O \rightarrow P, O \rightarrow P, K \rightarrow L$. Applying the same rule to DOOR: $D \rightarrow E, O \rightarrow P, O \rightarrow P, R \rightarrow S$. Therefore, DOOR = EPPS

- 4) Answer: 243

Explanation: Each number is multiplied by 3 to get the next number. $3 \times 3 = 9, 9 \times 3 = 27, 27 \times 3 = 81$, and $81 \times 3 = 243$.

5) Answer: All roses are flowers

Explanation: The first statement directly tells us that every rose is a flower. We cannot conclude that roses are red because only some flowers are said to be red.

6) Answer: Sister

Explanation: Anuj's grandfather's only son is Anuj's father. The daughter of Anuj's father is Anuj's sister. Therefore, the woman is Anuj's sister.

7) Answer: 37

Explanation: The differences are 3, 5, 7, and 9. The differences increase by 2 each time, so the next difference is 11. Therefore, $26 + 11 = 37$.

8) Answer: EPH

Explanation: Each letter is moved one position forward. $C \rightarrow D$, $A \rightarrow B$, $T \rightarrow U$. Therefore, $D \rightarrow E$, $O \rightarrow P$, $G \rightarrow H$, giving EPH

9) Answer: East

Explanation: He first moves 5 km north, then 3 km east, and finally 5 km south. The north and south movements cancel each other, leaving him 3 km east of his starting point.

10) Answer: All managers are employees

Explanation: The statement "all managers are employees" directly establishes that every manager is an employee. We cannot conclude that managers are graduates because only some employees are graduates.